



A Bio-Inspired Jump-Gliding Robot

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Motivation

Do something cool, something no one ever done before !
Always exploring !

Overview

Upper Platform

- Bearing
- Clutch
- Motor
- Connector

Jumping Module

- Carbon fiber rod
- Ultra-High-Molecular-Weight Polyethylene

Electrical Module

- Battery
- Arduino Nano
- Wires

Lower Platform

- Connection structure

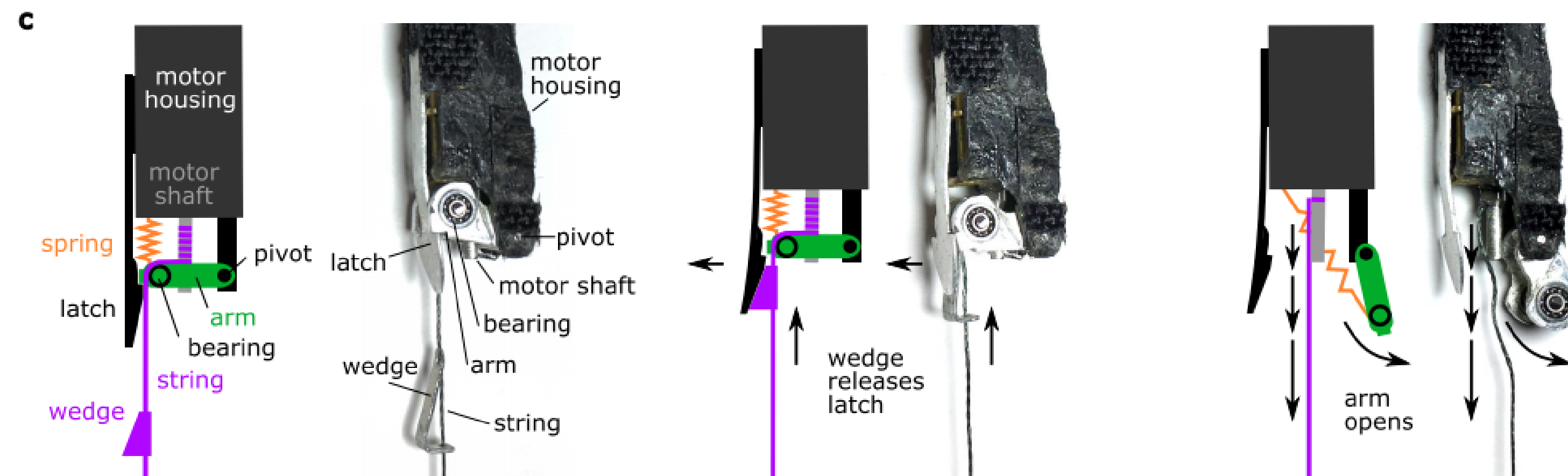
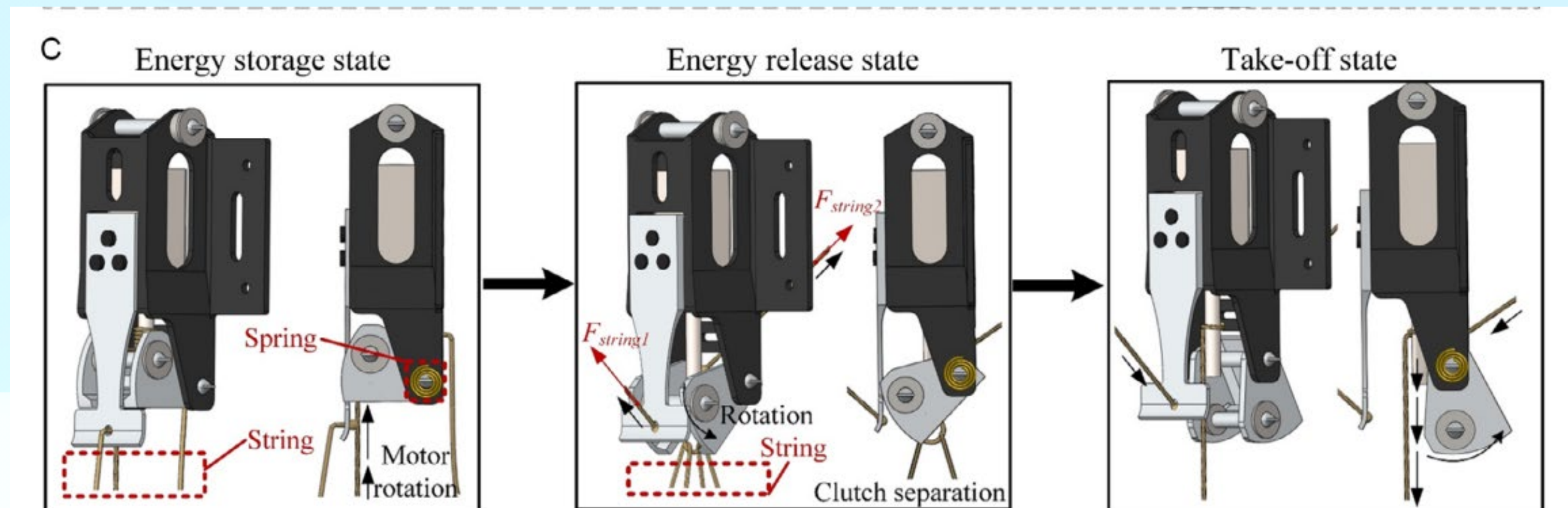


Jumping Moudule

- Release Mechanism
- Testing & Performance

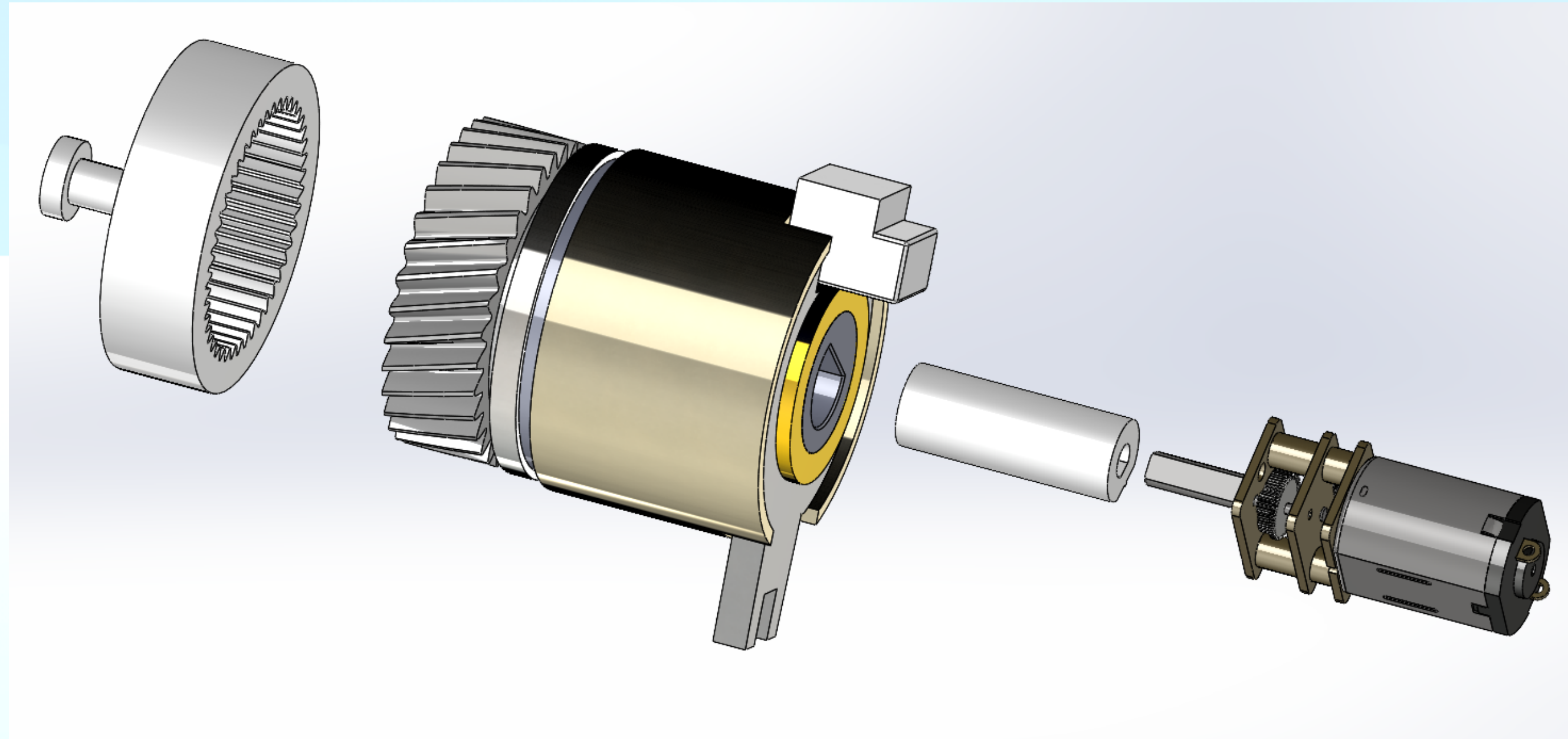
Release Mechanism

Others Only for One Time !



Release Mechanism

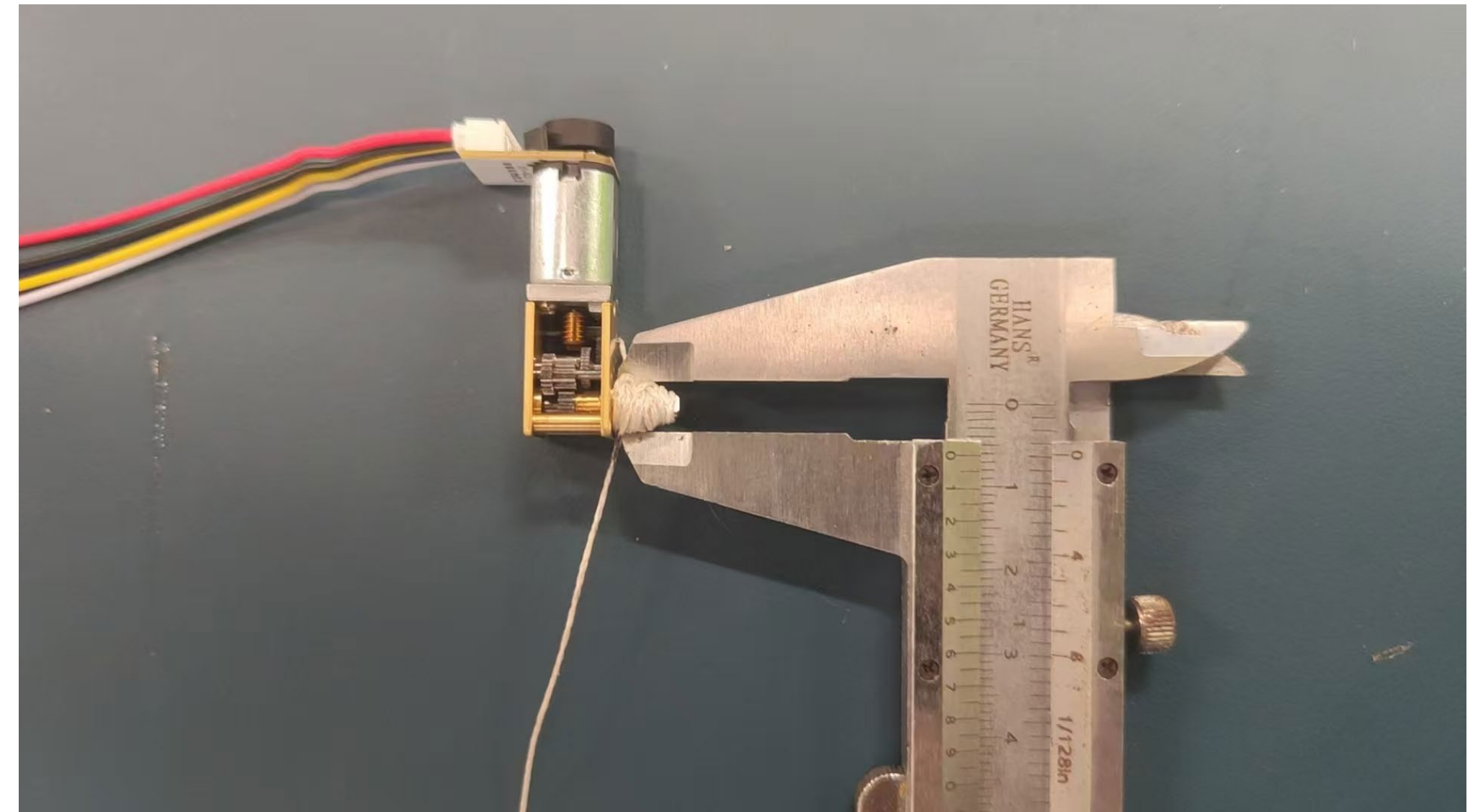
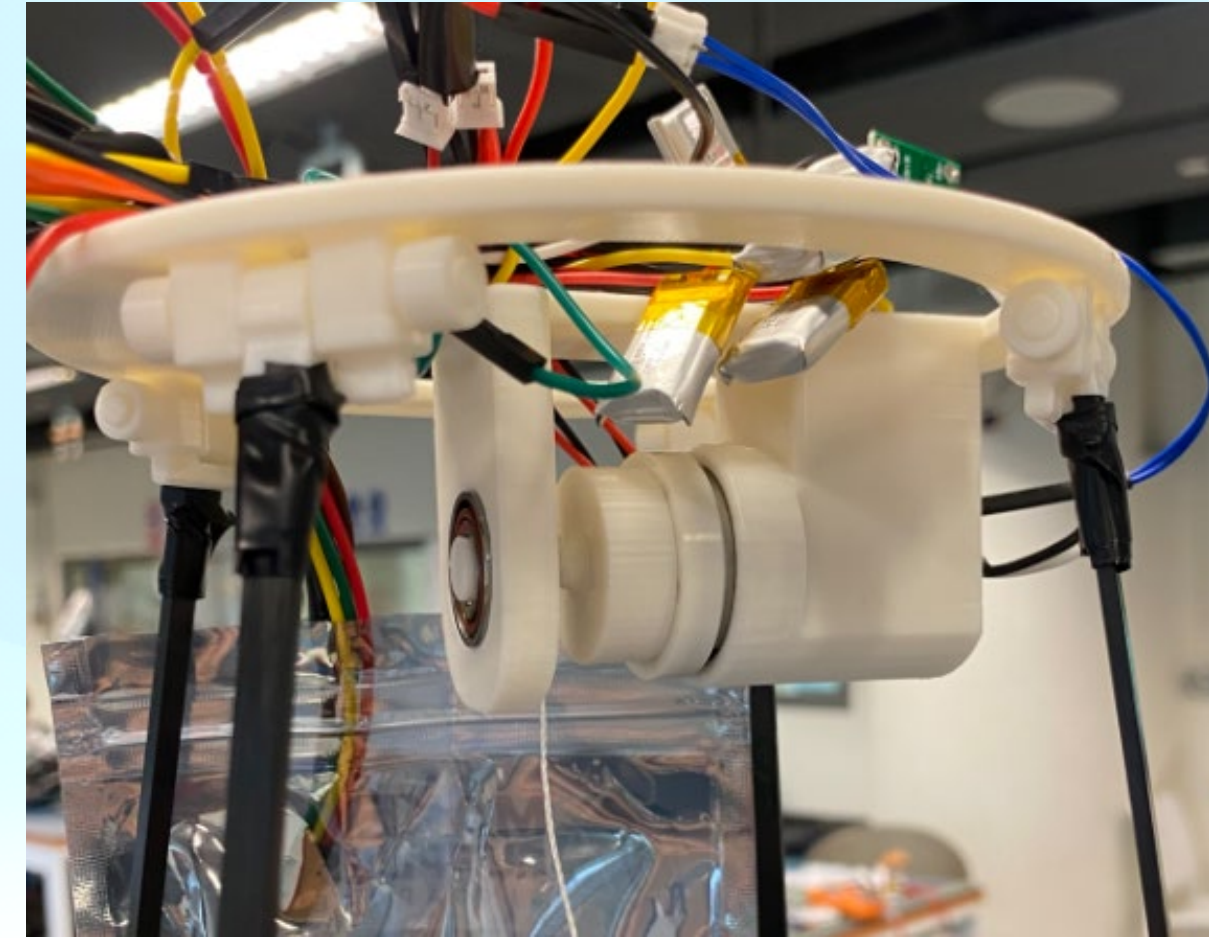
We can do many times !



1000:1 Micro Metal Gearmotor



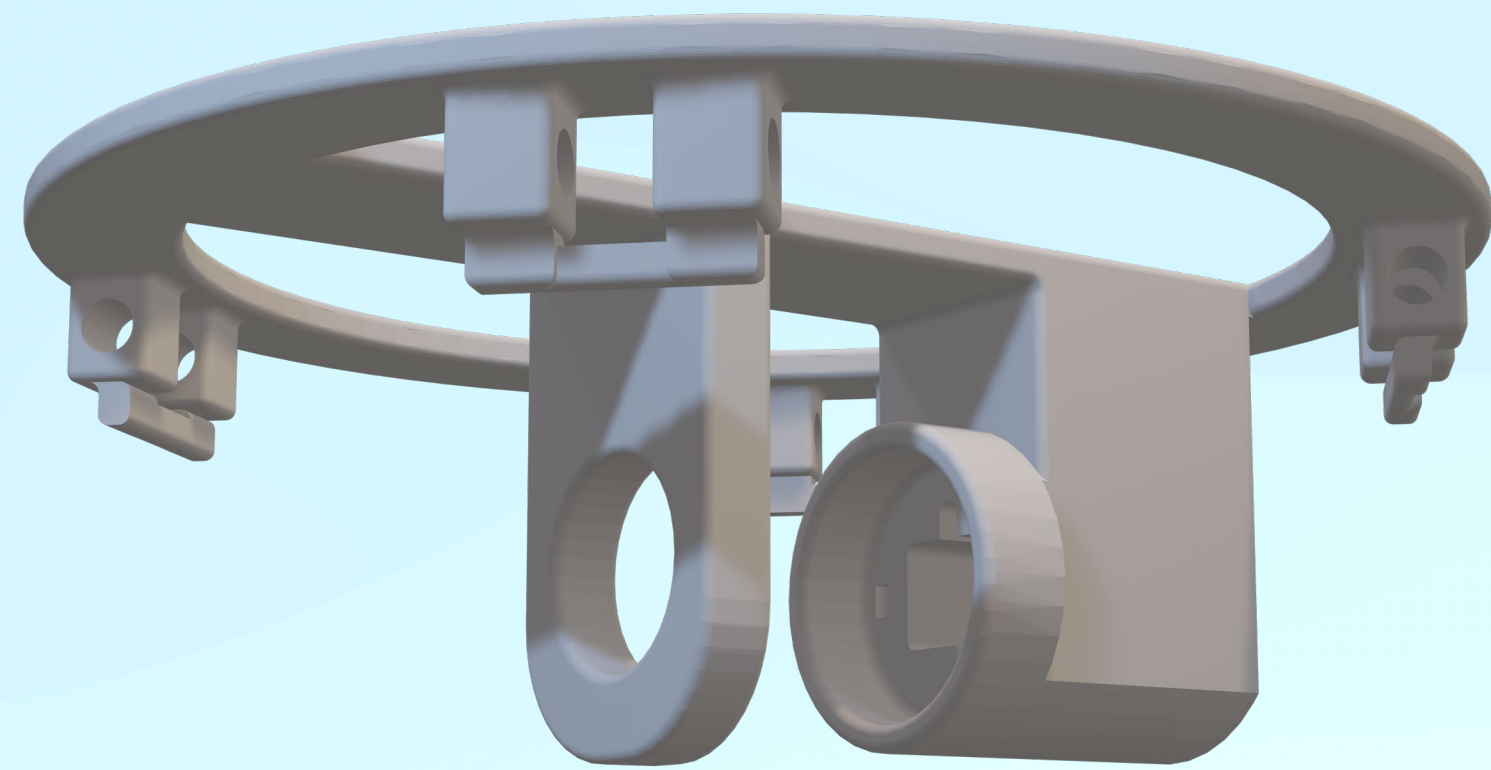
Release Mechanism



Iteration

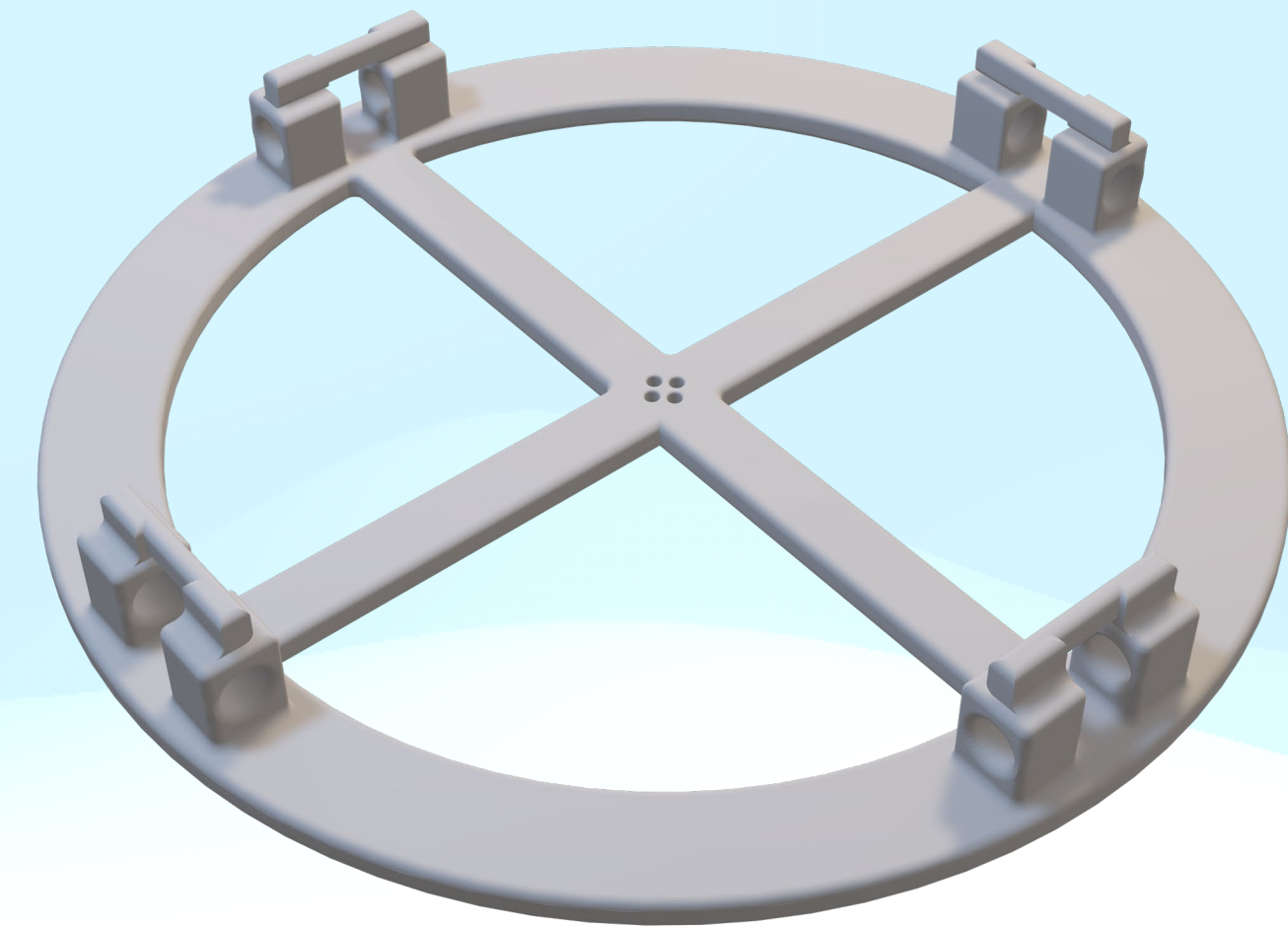


Release Mechanism



Upper Platform

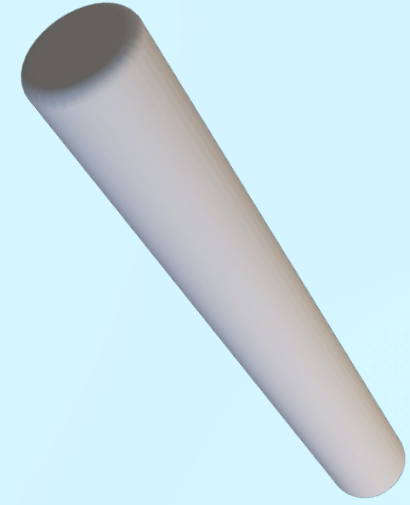
Designed to mount the motor, clutch, bearing connectors, and jumping mechanism.



Lower Platform

Designed to mount the jumping mechanism, together with the connectors on the upper platform, supports the tensioning lines used for energy storage.

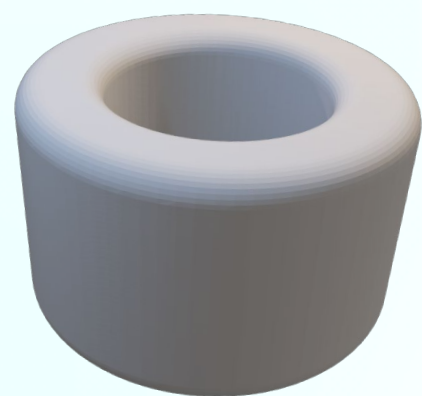
Release Mechanism



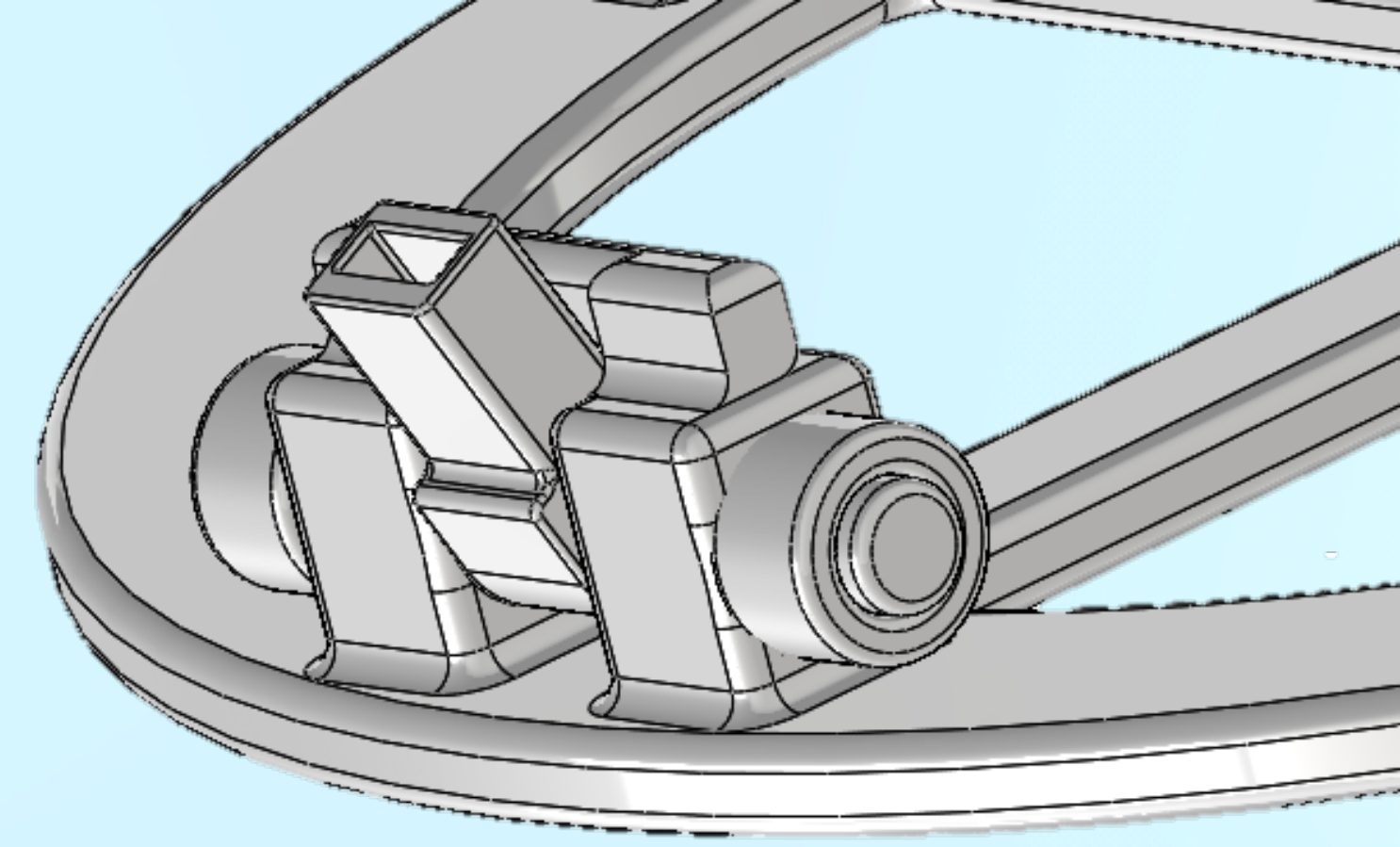
Length: 34mm
Diameter: 4.5mm



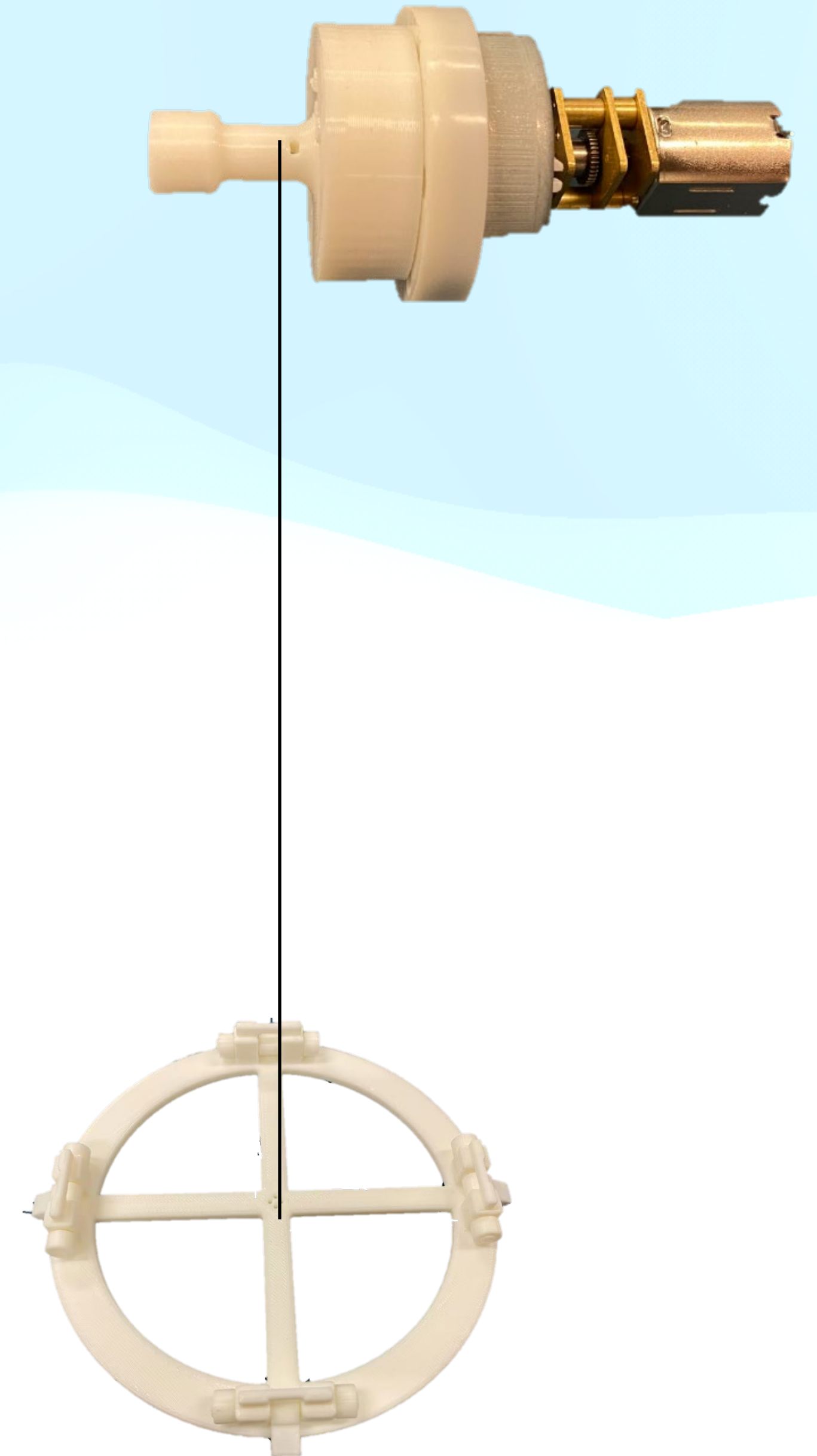
Groove: 2.1 x 5.1
Round hole: 4.8mm



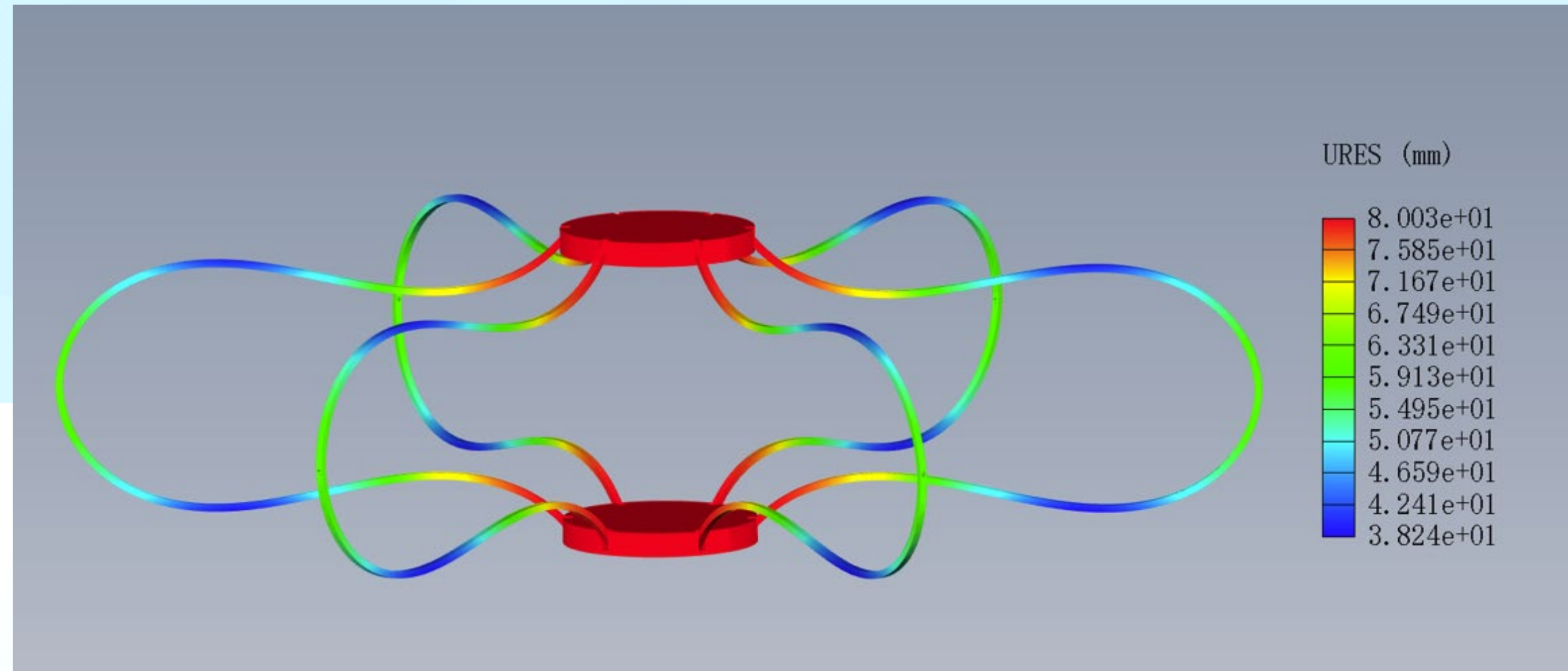
Inner diameter: 4.6mm
Outer diameter: 8mm



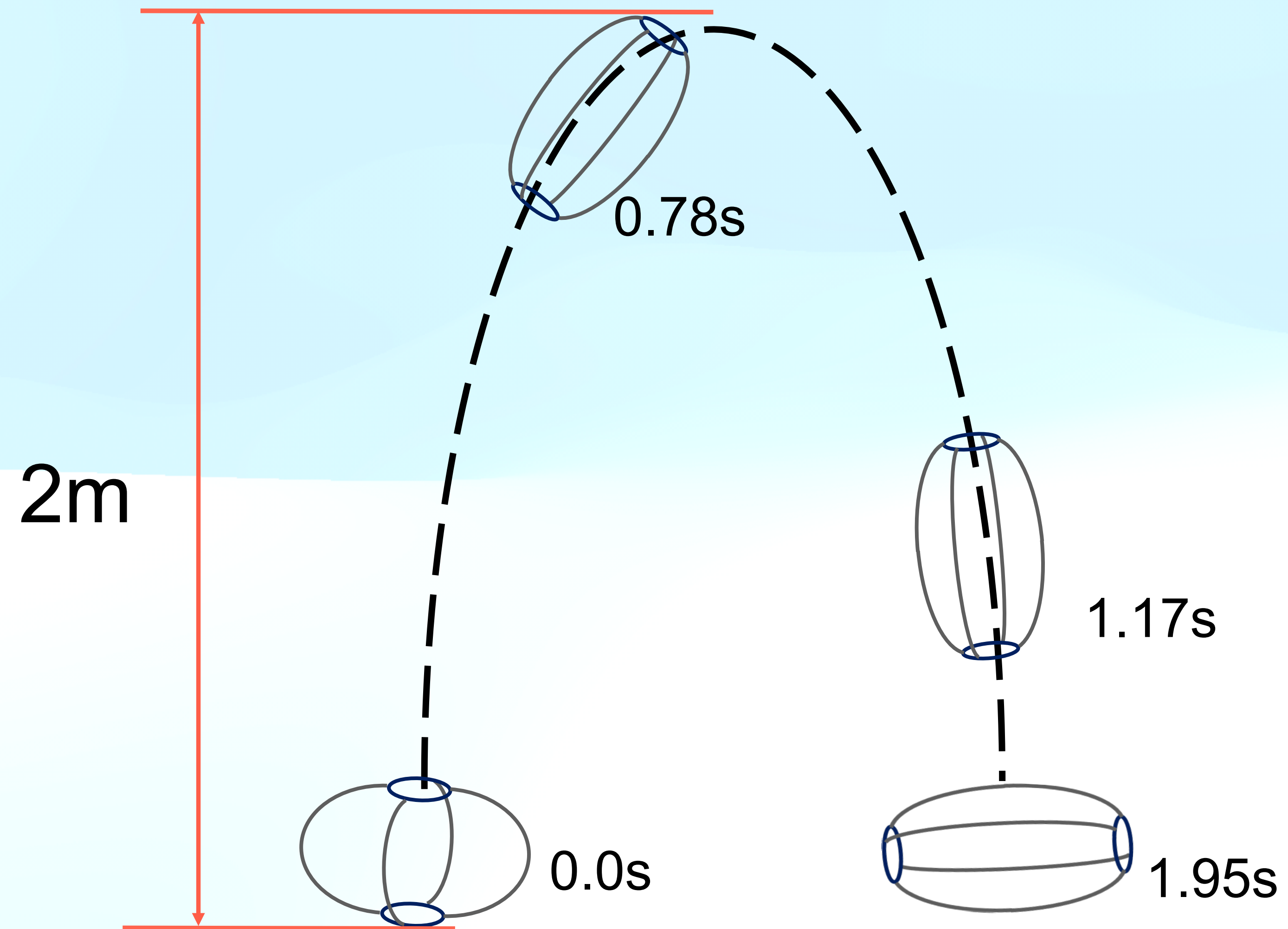
Release Mechanism



Testing & Performance

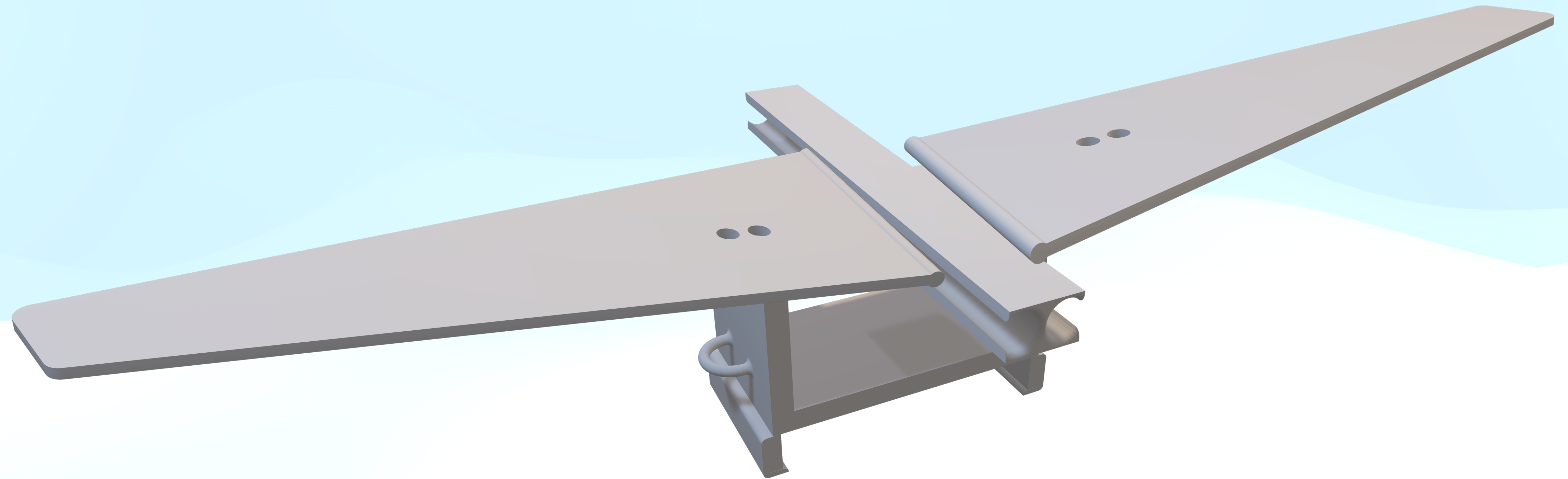


Testing & Performance





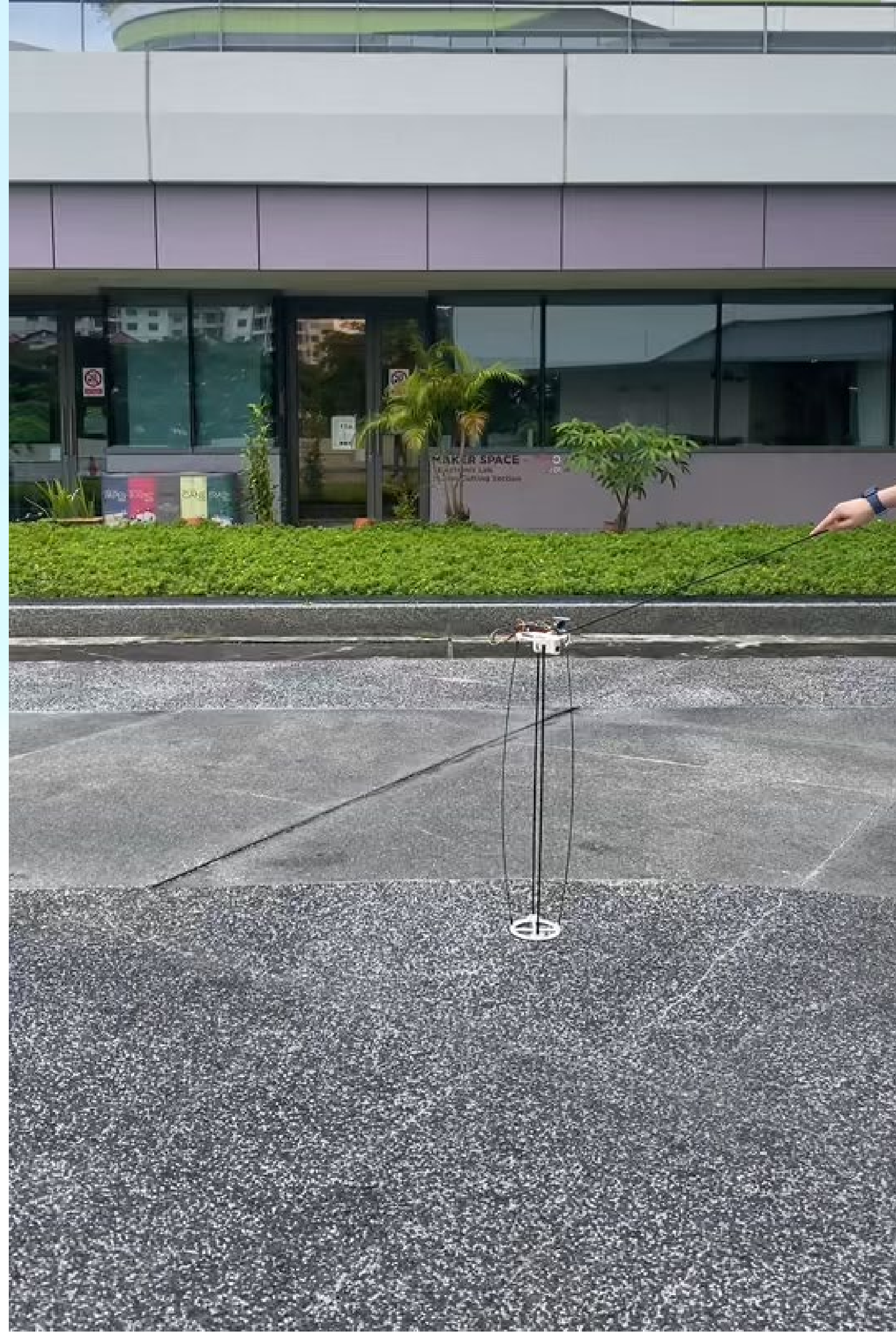
Gliding Module



Components	Mass [g]	Amount	Total Mass [g]
Carbon Fiber Jumper	64.0	1	64.0
EM Clutch	23.0	1	23.0
Servo Motor	16.0	1	16.0
Battery Pack	1.2	8	13.6
Arduino Nano 33 Ble Rev2	6.1	1	6.1
Electrical Wires	15	1	15
Gliding Wing	25.0	2	N.A
Connection Structure	35.3	4	35.3

$$M_{total} = 175 \text{ g}$$

Video





Summary

- **Redesigned Energy Storage System**
- **Custom Connectors**
- **Extensive Use of 3D Printing**
- **Lightweight Yet Strong**

Potential Application

- **Low-gravity extraterrestrial exploration**
- **Search-and-rescue in extreme environments**
- **Exploration in remote or dangerous areas**

Reference

- [1] Datar, C. Pan, M. Nazeri, and X. Xiao, “Toward wheeled mobility on vertically challenging terrain: Platforms, datasets, and algorithms,” arXiv Preprint, arXiv:2303.00998, Mar. 2023.
- [2] Y. Gao, B. Su, L. Jiang, and F. Gao, “Multi-legged robots: progress and challenges,” Natl. Sci. Rev., vol. 10, no. 5, p. nwac214, Oct. 2022, doi:10.1093/nsr/nwac214.
- [3] S. A. H. Mohsan, N. Q. H. Othman, Y. Li, M. H. Alsharif, and M. A. Khan, “Unmanned aerial vehicles (UAVs): practical aspects, applications, open challenges, security issues, and future trends,” Intell. Serv. Robot., vol. 16, no. 1, pp. 109–137, Jan. 2023, doi:10.1007/s11370-022-00452-4.
- [4] P. Liljebäck, K. Y. Pettersen, Ø. Stavdahl, and J. T. Gravdahl, “Hybrid modelling and control of obstacle-aided snake robot locomotion,” IEEE Trans. Robot., vol. 26, no. 5, pp. 781–799, Nov. 2010, doi:10.1109/TRO.2010.2056211.
- [5] A. A. Transeth, K. Y. Pettersen, and Ø. Stavdahl, “A survey on snake robot modeling and locomotion,” Robotica, vol. 27, no. 2, pp. 271–284, Mar. 2009, doi:10.1017/S0263574709005414.
- [6] A. Krishnan, J. J. Socha, P. P. Vlachos, and L. A. Barba, “Lift and wakes of flying snakes,” Phys. Fluids, vol. 26, no. 3, Art. no. 031901, Mar. 2014, doi:10.1063/1.4866444.

Reference

- [7]J. J. Socha, “Gliding Flight in Chrysopelea: Turning a Snake into a Wing,” *Integr. Comp. Biol.*, vol. 51, no. 6, pp. 969–982, Dec. 2011, doi:10.1093/icb/icr092.
- [8]E. W. Hawkes, C. Xiao, R.-A. Pelloquin, C. Keeley, M. R. Begley, M. T. Pope, and G. Niemeyer, “Engineered jumpers overcome biological limits via work multiplication,” *Nature*, vol. 604, no. 7907, pp. 657–662, Apr. 2022, doi:10.1038/s41586-022-04606-3.
- [9]less L, Rappaport NJ, Jacobson RA, Racioppa P, Stevenson DJ, Tortora P, Armstrong JW, Asmar SW. Gravity field, shape, and moment of inertia of Titan. *Science*. 2010 Mar 12;327(5971):1367-9. doi: 10.1126/science.1182583. PMID: 20223984.
- [10]Fulchignoni M, Ferri F, Angrilli F, Ball AJ, Bar-Nun A, Barucci MA, Bettanini C, Bianchini G, Borucki W, Colombatti G, Coradini M, Coustenis A, Debei S, Falkner P, Fanti G, Flamini E, Gaborit V, Grard R, Hamelin M, Harri AM, Hathi B, Jernej I, Leese MR, Lehto A, Lion Stoppato PF, López-Moreno JJ, Mäkinen T, McDonnell JA, McKay CP, Molina-Cuberos G, Neubauer FM, Pirronello V, Rodrigo R, Saggin B, Schwingenschuh K, Seiff A, Simões F, Svedhem H, Tokano T, Towner MC, Trautner R,

Reference

Withers P, Zarnecki JC. In situ measurements of the physical characteristics of Titan's environment.

Nature. 2005 Dec 8;438(7069):785-91. doi: 10.1038/nature04314. Epub 2005 Nov 30. PMID: 16319827

[11]Won Dong Shin et al 2019 Bioinspir. Biomim.14 056009

[12]F. Zhao, W. Wang, J. Zhang, J. Wyrwa, and F. Sun, "Aerodynamic Characteristics and Pitching

Adjusting Mechanism of the Flying Squirrel With Deployed Patagium," IEEE Access, vol. 7, pp. 185554–

185565, 2019, doi: 10.1109/ACCESS.2019.2961451. Demonstration of a Flying-Squirrel-Inspired Jumping

Robot with Two Modes," Applied Sciences, vol. 11, no. 8, p. 3362, 2021, doi: 10.3390/app11083362.



Thank you !